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[Home](#) > [Topics A-Z](#) > [Nipah virus disease](#)

> Disease information on Nipah virus disease

Disease information on Nipah virus disease

Factsheet

Nipah virus (NiV) is an emerging zoonotic virus that can be transmitted to humans from animals (both wild and domestic). Fruit bats of the *Pteropodidae* family are the natural hosts of Nipah virus. NiV infection can be asymptomatic or mild (subclinical), but it also causes acute respiratory illness and fatal encephalitis. The case fatality rate is estimated between 40–75% or higher, and it varies depending on the specific NiV strain [1], access to and quality of healthcare, or the severity of illness [2]. There are no specifically approved treatments or vaccines available for NiV infection, either for people or animals.

So far, NiV outbreaks have only been described in Asia. The first reported outbreak occurred in 1998–1999 and involved pig farmers in the village of Sungai Nipah (Malaysia) [3]. Since then, sporadic outbreaks have been documented in Bangladesh, India, the Philippines, and Singapore [4-7].

NiV infection is one of the priority diseases on the World Health Organization (WHO) R&D Blueprint because of its epidemic or pandemic potential [8].

Case definition

NiV infection is not a notifiable disease at the EU/EEA level and there is no EU case definition.

However, NiV infection case definitions have been proposed in scientific literature [9] and by national public health institutes in various countries (India [10], Malaysia [11]).

The pathogen

NiV belongs to the *Henipavirus* genus within the [*Paramyxoviridae*](#)  family.

The *Paramyxoviridae* family consists of enveloped, non-segmented, negative-sense RNA viruses. Their genomes code for six structural proteins: nucleocapsid (N), phosphoprotein (P), matrix protein (M), fusion protein (F), glycoprotein or receptor-binding protein (G), and large protein or RNA polymerase (L).

Many paramyxoviruses are host-specific, and several are pathogenic to humans, including NiV, measles virus, mumps virus, Hendra virus (HeV), and several parainfluenza viruses. NiV has a broad host range that includes humans, bats, pigs, sheep, goats, dogs, cats, and horses.

The [*genus Henipavirus*](#)  has five recognised species (*Cedar henipavirus*, *Ghanaian bat henipavirus*, *Hendra henipavirus*, *Mojiang henipavirus*, and *Nipah henipavirus*) that are distinguished on the basis of the amino acid sequences of their large protein (L) [12]. Species demarcation is based on the distance in phylogenetic analysis of complete L protein amino acid sequences. Phylogenetic tree topology and branch length (0.03) between the nearest node and the tip of the branch determines whether two genotypes belong to the same species [12].

Recently, further henipaviruses were discovered in South-East Asia in small mammals [13,14]. Metagenomic and serological studies indicate human infections in China with a henipavirus tentatively named Langya virus [15].

Based on sequence differences of either whole or partial nucleocapsid (N) gene, NiV strains are divided into two clades (Clade I: NiV-B in Bangladesh and India, and Clade II: NiV-M in Malaysia) which suggests two separate NiV introductions in these countries [16,17]. The two strains have >92% genetic identity and amino acid homology, but clinical features and transmission routes differ in India/Bangladesh and Malaysia [3,18]. Phylogenetic analysis also confirmed two distinct NiV-M subclades circulating in northern and southern Malaysia; these two subclades likely originate from distinct hosts [19].

In the first stage of illness, virus detection can be performed in the epithelial cells of the bronchioles. In experimental animal models, viral antigens can be detected in the bronchi and alveoli. In the later stages of illness, virus can also be found in the lung endothelia and the blood

either freely or in host leukocyte-bound form. Eventually, lungs, spleen, kidneys and brain are targeted by the virus and the patient suffers from multiple organ failure [3]. Ephrin-B2 (EFNB2) and ephrin-B3 (EFNB3) were identified as the main functional receptors for both NiV clades. The presence of these receptors in many vertebrates can explain the wide host range of henipaviruses [20]. In addition, the fact that these receptors are often located on the arterial endothelial cells can explain the systematic nature of the infections [20].

NiV has been classified as a Risk Group 4 pathogen, which means specimens containing NiV need to be handled using special containment and barrier protection measures. This applies to laboratory personnel handling samples and healthcare professionals caring for potentially infected patients, as well as people taking care of dead bodies. While diagnostic RT-PCR and serology on inactivated samples can be done under biosafety level (BSL)-2 conditions, virus inactivation has to be done at BSL-3. Virus propagation, isolation, quantification and neutralisation require BSL-4 facilities.

NiV is relatively stable, and it can survive for up to three days in some fruit juices or fruits at local environmental temperatures. It can remain infectious for at least seven days in date palm sap kept at 22°C. The virus has a half-life of 18 hours in the urine of fruit bats. It remains viable at 70°C for one hour (only the viral concentration decreases). However, it can be completely inactivated by heating at 100°C for more than 15 minutes. Given its enveloped nature, NiV can be inactivated by soaps, detergents and sodium hypochlorite (10 000 ppm) [3,21]. The use of ultraviolet-C and medium-base lights was found to be an effective way to reduce NiV infectivity in serum and plasma [22], with the combination of heating [23]. More information on inactivation methods for NiV-infected material is available in Widderspick et al [24].

Clinical features and sequelae

Up to 11% of infected individuals remain asymptomatic and the proportion of asymptomatic cases seems to depend on the specific NiV strain [25]. Symptomatic individuals may present mild respiratory symptoms to severe respiratory distress or encephalitis. People infected with the NiV-B strain are more likely to develop atypical pneumonia and severe respiratory problems, including acute respiratory distress [26].

Initial symptoms of NiV infection may include fever, headache, myalgia (muscle pain), nausea, vomiting, sore throat, cough and/or respiratory discomfort. Acute encephalitis might develop with the following

symptoms: dizziness, drowsiness, altered state of consciousness and any other indicative neurological signs. In severe cases, encephalitis and seizures occur and may lead to coma within 24–48 hours. Progression to encephalitis indicates a poor prognosis, with death within a median of six days after the onset of symptoms [27].

Most people who survive encephalitis make a full recovery, but neurological dysfunctions remain in around 20% of survivors. These include recurrent seizures, overwhelming fatigue and altered behaviour. A small proportion of patients can have relapses or delayed onset of encephalitis (weeks, months or even years post-recovery) with fatal outcomes [19,28-30].

The case fatality rate usually ranges between 40–75% but can be higher depending on the NiV strain, the severity of the disease and the availability of adequate and high-quality healthcare facilities [2,17,31].

NiV can infect and cause subclinical or clinical diseases in domestic animals such as dogs, cats, pigs, horses, goats and sheep [32]. The illness is similar to the human disease i.e. neurological or respiratory. It seems to be mostly acute and self-limiting, but can also be fatal [33].

Epidemiology

NiV has caused a few known outbreaks in Asia. It emerged in 1998 during an outbreak among pig farmers in the village of Sungai Nipah in Malaysia [3]. In March 1999, there was another outbreak in Singapore including 11 abattoir workers linked to imported pigs from a farm in Malaysia which were infected by NiV [7]. In total, 246 cases were reported from these two outbreaks. Pig farmers and abattoir workers were identified as the high-risk group. A period of widespread forest burning for agricultural use, which sparked the migration of fruit bats from Indonesia to other parts of South-East Asia was considered to be an underlying reason for the geographical expansion of NiV reservoir hosts [34].

Since the initial detection, three countries described single or sporadic reoccurring outbreaks: Bangladesh (the first outbreak was in 2001, and then there were nine subsequent outbreaks till 2010 and 2011), India (2001, 2007, 2018, 2019, 2021) and the Philippines in 2014. In Bangladesh and India, the most likely sources of infection were fruits or fruit products (such as raw date palm sap) contaminated with urine or saliva from infected fruit bats [27]. Human-to-human transmission was also reported [17,35]. In the Philippines, the most commonly described

routes of transmission to humans were direct exposure to infected horses, contact with contaminated materials during the slaughtering of sick horses, or consumption of raw or undercooked meat from infected horses [6].

Anti-NiV or cross-reacting anti-NiV antibodies have been detected in fruit bats from Bangladesh, Cambodia, China, Indonesia, India, Madagascar, Malaysia, New Caledonia, Papua New Guinea, Thailand and Vietnam [3,18]. NiV RNA was detected in bats in Timor-Leste (both Pteropid and non-Pteropid), Bangladesh, India and Thailand. Moreover, Ghanaian bat henipavirus RNA (a species of the *Henipavirus* genus) was identified in Ghana [36].

There are many unknown aspects about the patterns of NiV circulation in bats and the factors which may cause the virus to spillover to domestic animals. The role of bats other than Pteropid bats in the dissemination of NiV remains unclear. The only fruit bat residing in Europe is the Egyptian fruit bat (*Rousettus aegyptiacus*), which does not support NiV replication experimentally [37]. Henipavirus antibodies were found in China in Daubenton's bat (*Myotis daubentonii*) – a species of insectivorous bats widely found in Europe and parts of Asia. However no specific RNA was detected [38].

Virus shedding by bats is cyclical, driven by demographic factors (population turnover, migration of infected bats) and immunological factors (waning protection, stress during pregnancy) [39]. No seasonal pattern in human infections could be identified besides the seasonal consumption of date palm sap (contaminated with urine or saliva from infected fruit bats) in Bangladesh during winter months [40].

Transmission

The incubation period of NiV generally ranges between 4–21 days [2]; however, longer incubation periods have also been observed, notably during the outbreak in Malaysia when the incubation period was up to two months. It is presumed that infected persons remain infectious till 21 days after the onset of symptoms.

NiV is transmitted either via direct contact with infected people or animals or their body fluids, or via consumption or handling of contaminated products. Mother-to-child transmissions have not been recorded [26]. However, the possibility was demonstrated in an animal experiment [41].

NiV can be transmitted to animals by:

- The consumption of contaminated fruit, with saliva or urine from infected fruit bats [3] or the consumption of meat from an infected animal [6].
- Direct contact with other infected animals [42]. Rectal shedding of viable virus was reported in animal models [43].

NiV can be transmitted to people by:

- Direct contact with infected wild or domesticated animals, or their excretions and secretions (such as urine, saliva, blood, pharyngeal and respiratory secretions). Transmission to humans via virus-containing aerosols should also be considered [44].
- Consumption of unprocessed meat from infected animals, or unheated/raw food products that have been contaminated by body fluids of infected animals (such as palm sap or fruit contaminated by an infected fruit bat).
- Contact with the body fluids of an infected person (such as nasal or respiratory droplets, urine, blood).

The main mode of transmission varied between the affected countries [18]. In Malaysia and Singapore, the outbreaks were linked to pigs as intermediate hosts, while in the Philippines horses were the intermediate hosts [6,7,18,45]. In all instances, the consumption of fruits contaminated with bat droppings is believed to have caused the initial infections in domestic animals. In India and in Bangladesh, outbreaks were linked to the consumption of date palm sap [18]. Human-to-human transmissions in the community and/or hospital settings were reported in several countries including Bangladesh, India and the Philippines [6,35,46-49].

Diagnostics

The initial signs and symptoms of NiV infection are nonspecific, which is why it is often misdiagnosed at first. A syndromic testing approach for pathogens causing symptoms such as febrile, respiratory or encephalitic disease could be considered for more accurate diagnosis [19]. Moreover, HeV and NiV are closely related, and diagnostic assays can cross-react depending on the RNA or antigen targeted.

In the early stages of disease, NiV infection can be confirmed by RT-PCR testing of RNA extracted from throat and nasal swabs, cerebrospinal fluid, urine, and blood. Animal experiments demonstrated shedding in faeces, but stools are not listed in humans amongst the standard

diagnostic samples. Exact virus kinetics are not known as well. Several narrow- and broad-range RT-PCR assays exist for strain detection, including a few commercial ones [19].

Later (10–14 days after the onset of symptoms) in the course of illness and after recovery, the diagnosis relies on antibody testing, using immunoassays (e.g. enzyme-linked immunosorbent assay (ELISA), and Luminex) or virus neutralisation. This also depends on the post-symptoms timing, the kinetics of antibodies and available laboratory facilities.

Virus isolation and virus neutralisation tests need to be performed at BSL-4 [3]. Neutralisation tests can also be performed at BSL-2 if pseudoviruses are used [3]. BSL-2 facilities can be used for RNA isolation on inactivated samples from suspected clinical materials, but proper risk assessments need to be observed for any work. Currently there are no points of care/rapid tests available for NiV [19].

Various diagnostic tests are available for animals.

Immunohistochemistry can be performed on infected animal or human tissues using specific anti-henipavirus antibodies. This is sometimes useful to confirm the diagnosis, although detection of viral RNA by RT-PCR and confirmation by sequencing are preferred methods.

Case management and treatment

Guidelines for the clinical management of patients with NiV infection have been made available in various countries, including affected countries such as Bangladesh, India and Malaysia. NiV clinical management is based on general, supportive treatment (ensuring fluid and electrolyte balance, oxygen inhalation if required), symptomatic treatment (treating fever, convulsions, shock), and adopting procedures to prevent further spread of the disease (implementing isolation measures, barrier nursing, safe handling of deceased bodies).

NiV-specific immunotherapeutic treatments (such as m102.4 monoclonal antibody against the G protein) are under development and evaluation. This treatment showed effectiveness in animal models [50]. Based on a completed phase 1 clinical trial, it can also be used in patients on a compassionate use basis.

As for treatment, several antiviral medications have been employed/tested for NiV infection. Remdesivir was found effective against NiV-B in non-human primates as a post-exposure prophylaxis when given 24 hours after intranasal and intratracheal lethal challenge

and treatment pursued for 12 days [51]. It may be complementary to immunotherapeutic treatments. In Malaysia, ribavirin was used to treat a small proportion of patients, but its efficacy in people remained unclear [52]. After this, it was also used in India with supposed positive outcomes [53]. However, governing conditions and the true efficacy of this treatment remain unclear [54]. Other drugs tested and showing inhibition either in vitro or in animal models are favipiravir (T-705) [55] and 4'-chloromethyl-2'-deoxy-2'-fluorocytidine (ALS-8112) [56].

Public health control measures

Raising awareness about the risk factors and mitigating measures are key to reducing the risk of NiV exposure and transmission. The present options to prevent secondary transmissions are active case finding, contact tracing, isolation and quarantine of cases and their contacts. Human-to-human transmission can be reduced by avoiding close contact with infected individuals, using protective equipment, and following good hand hygiene.

NiV transmission may have happened through different routes depending on the epidemiological context in the country or region. Control measures should therefore be adapted to the transmission pattern. In Malaysia, culling of infected domestic animals and implementing disinfection measures were successful in stopping the outbreak [45]. Furthermore, infection prevention in livestock is an effective method in regions where they serve as intermediate hosts. Strategies involve keeping fruit trees (on which bats usually roost) away from livestock susceptible to NiV infection [6]. In regions in Bangladesh and eastern India particularly, it is essential to create public awareness about the need to boil fresh palm sap and the importance of preventing access of bats to the containers in which the sap is stored. In regions in south India, where transmission is from infected fruits, proper washing and peeling of vegetables and is essential to remove traces of bat excreta from them.

Infection control, personal protection and prevention

As symptoms during the prodromal phase are not specific, the early diagnosis of NiV infection can be challenging. Infection control measures and other preventive steps like hand hygiene, barrier nursing, safe handling of deceased bodies help to limit viral spread both in households and hospitals. Personal protection devices like masks

(providing sufficient filtration against airborne diseases), glasses and gloves, should also be used properly [3,26].

There is very limited data available worldwide about NiV disease. There has been no evidence of increased incidence rates or increased severity of the disease in specific population groups, such as pregnant women, infants or immunocompromised subjects. Length of exposure and contact with body fluids were increasingly associated with risk of infection [26]. In Malaysia, high-risk populations were people who handle pigs, pork sellers, pig cullers and so on [45]. Isolation of NiV from urine and faeces collected from the cages of animals infected with NiV-M and NiV-B strains suggests that environmental contamination could pose a risk of infection [43]. This also points to the danger of transmission in zoos when bats which have the potential to carry NiV are housed on the premises.

Currently no NiV vaccine is approved either for humans or animals. However, in Australia an HeV vaccine for horses is available, which cross-protects against NiV infection.

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